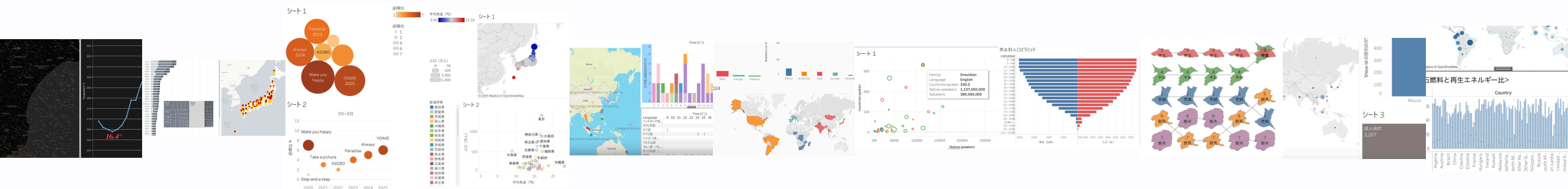


Gallery walk



0598

[Padlet link](#)

Language of Data Visualization

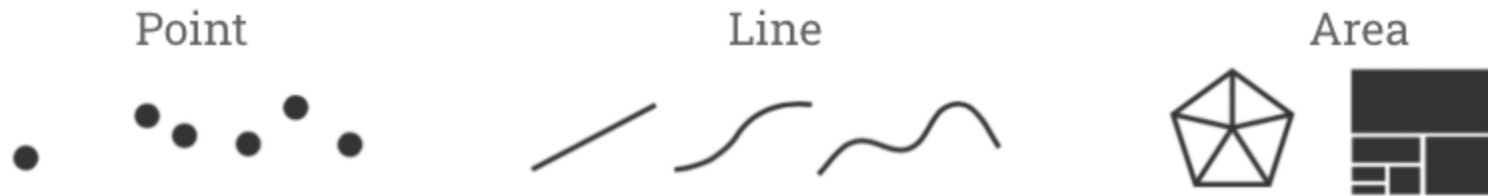
Marks and Channels



Marks • マーク

The shapes that represent data.

データを表す形

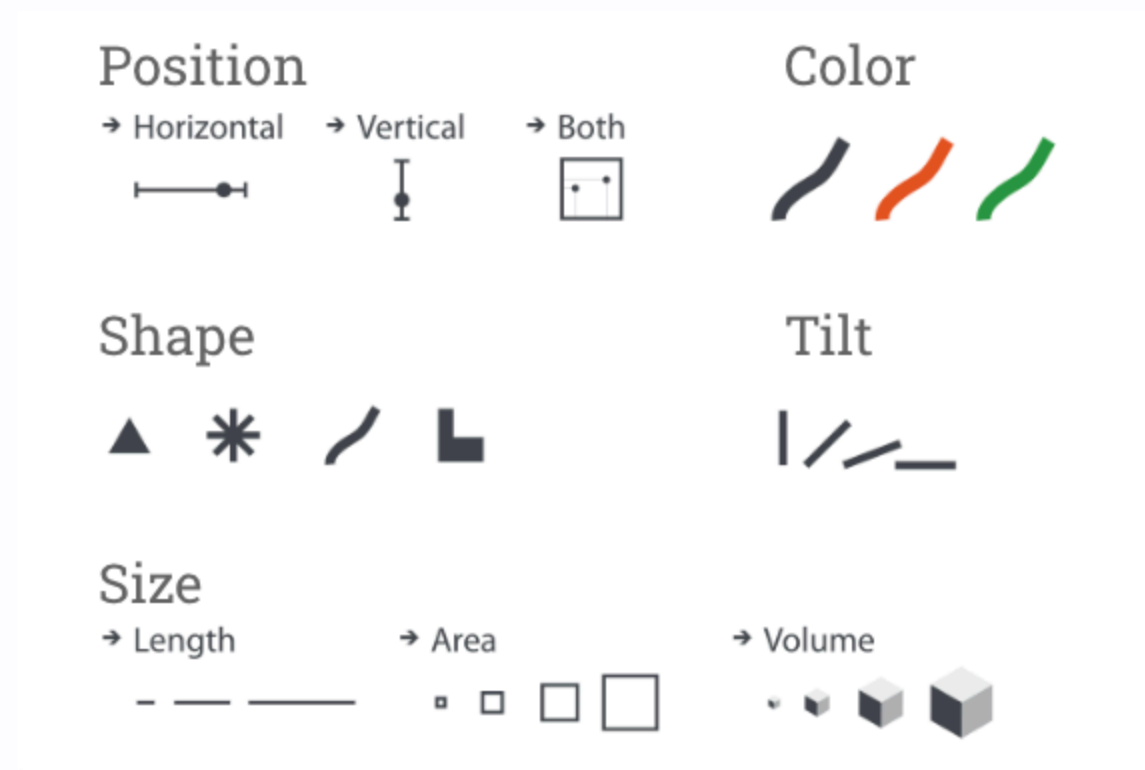


Graphics from Munzner. Visualization Analysis and Design (2015)

Channels ・ チャネル

Ways to vary the appearance of marks

マークの見た目を変える方法



Marks ▾

○ Circle ▾

 Color

 Size

 Label

 Detail

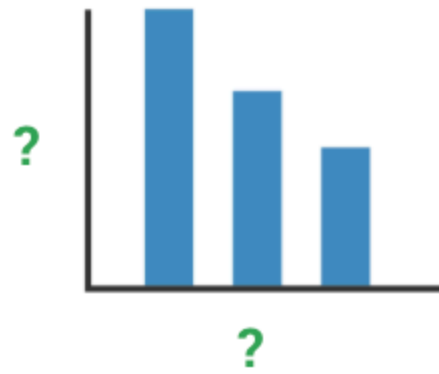
 Tooltip

 Region

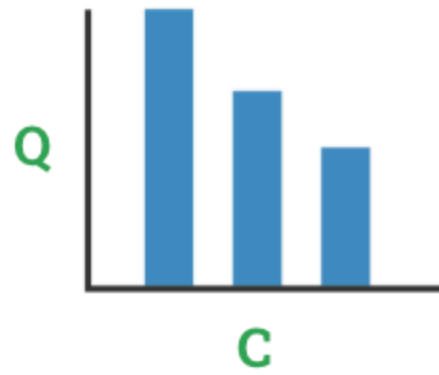
 SUM(Populatio..

 Country ▾

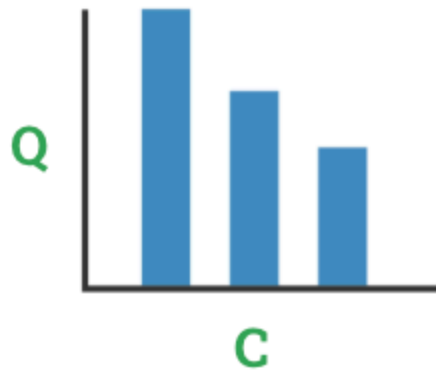
 Country



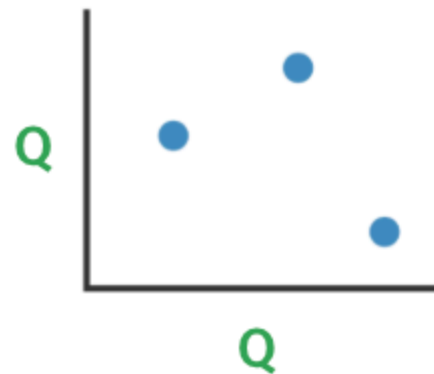
Mark: ?



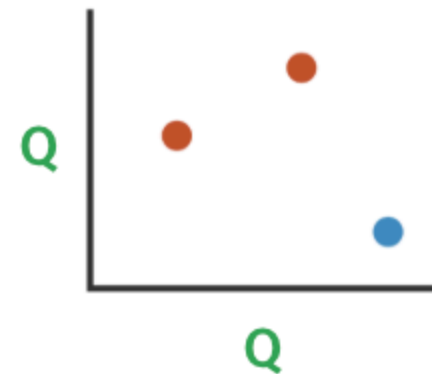
Mark: **line**



Mark: **line**



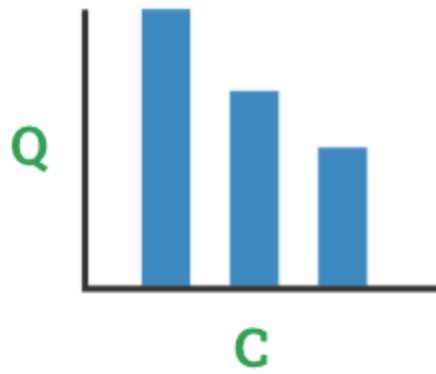
Mark: **point**



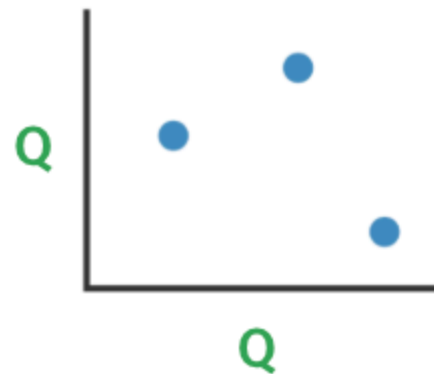
Mark: **point**

Channels:

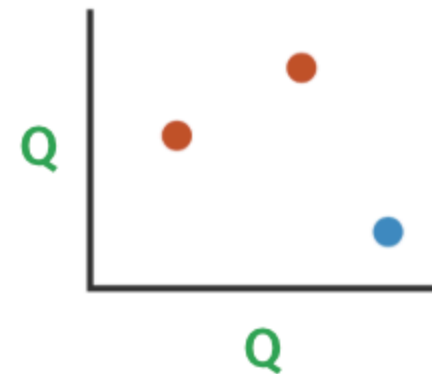
Color: **C**



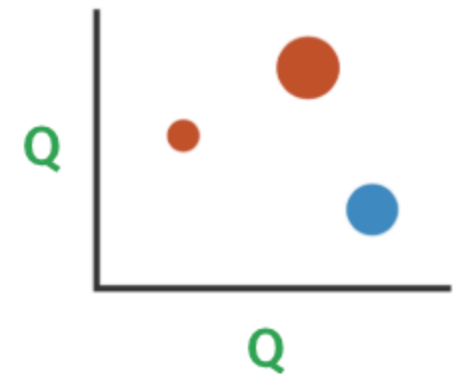
Mark: **line**



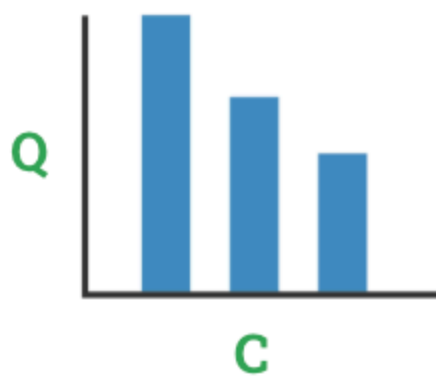
Mark: **point**



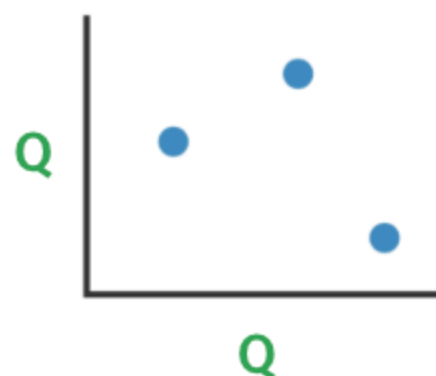
Mark: **point**
Channels:
Color: **C**



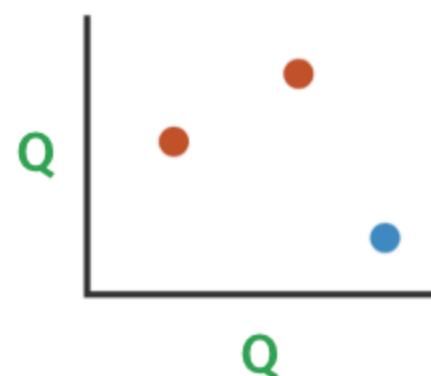
Mark: **point**
Channels: **?**



Mark: **line**



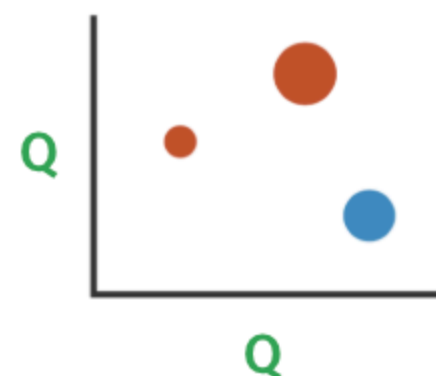
Mark: **point**



Mark: **point**

Channels:

Color: **C**



Mark: **point**

Channels:

Color: **C**

Size: **Q**

Position

→ Horizontal



→ Vertical



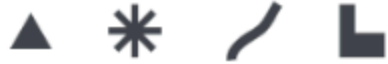
→ Both



Color



Shape



Tilt



Size

→ Length



→ Area



→ Volume



Position



Color



Categorical



Shape



Tilt

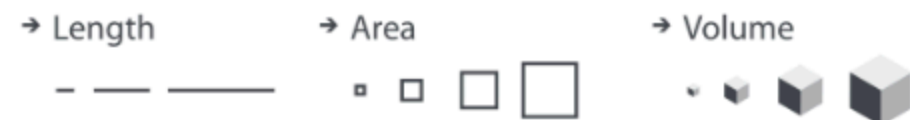


×

Ordinal



Size



Quantitative



Position

→ Horizontal



→ Vertical



→ Both



Color



Shape



Tilt



×

Categorical



Ordinal



Size

→ Length



→ Area



→ Volume



Quantitative



Filters

Marks

Circle

Color

Size

Label

Detail

Tooltip

SUM(Population)

SUM(Population)

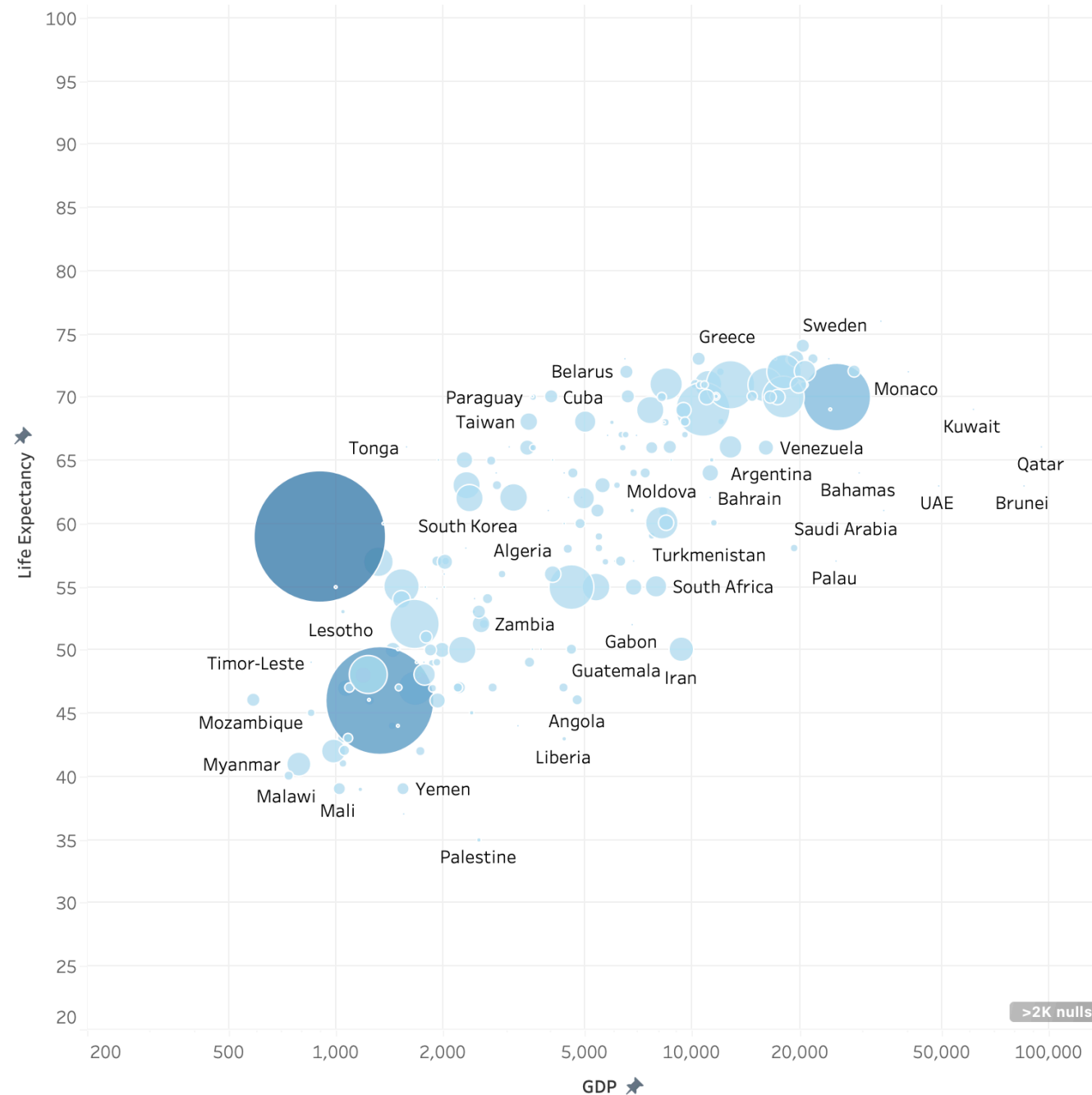
Country

Country

SUM(Population)



Hans - 1967



Position



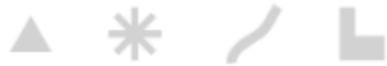
Color



Categorical



Shape



Tilt

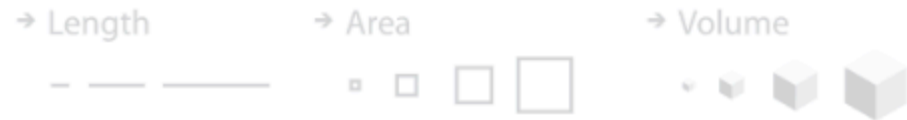


×

Ordinal



Size



Quantitative



Filters

Marks

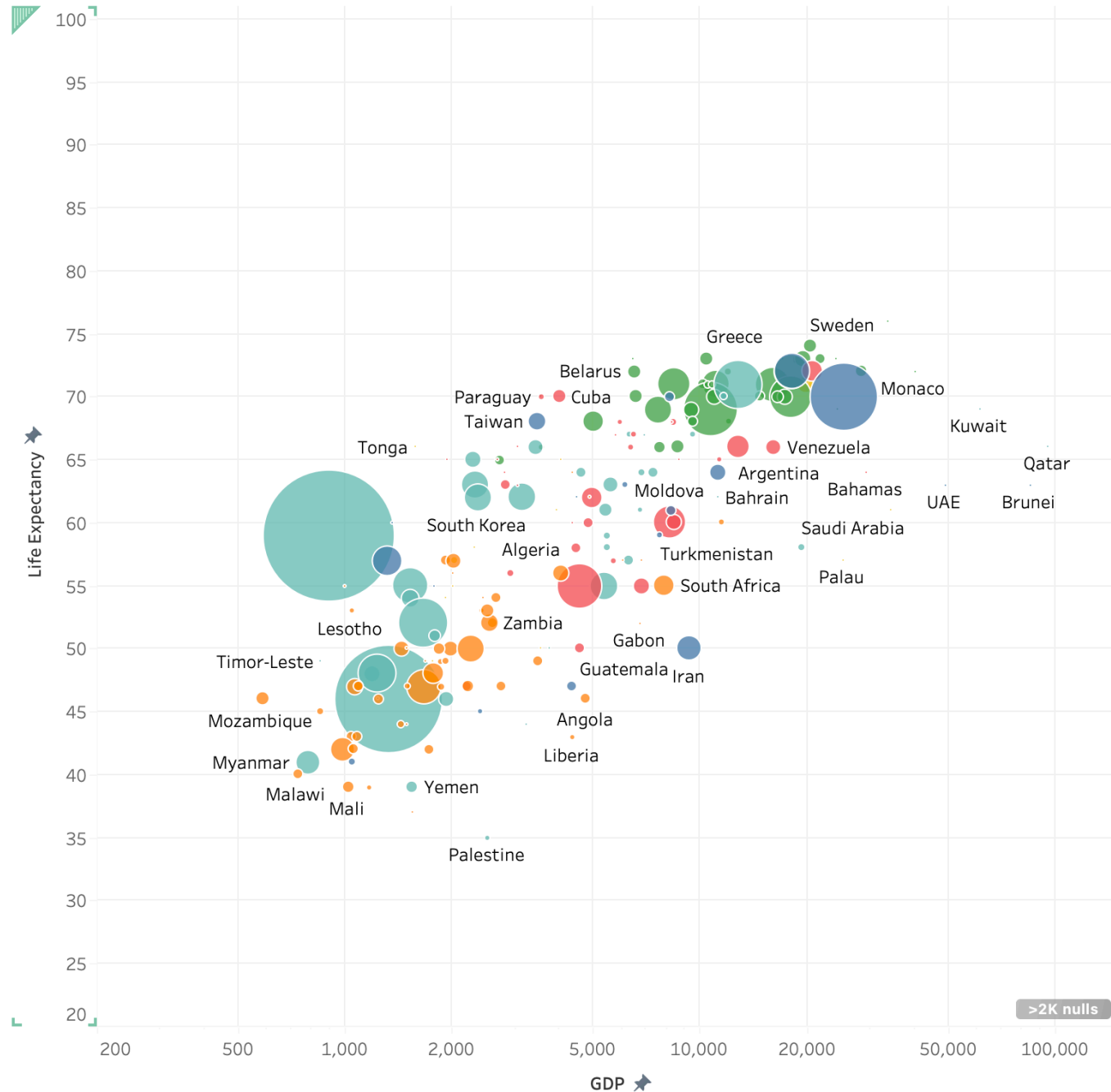
Circle

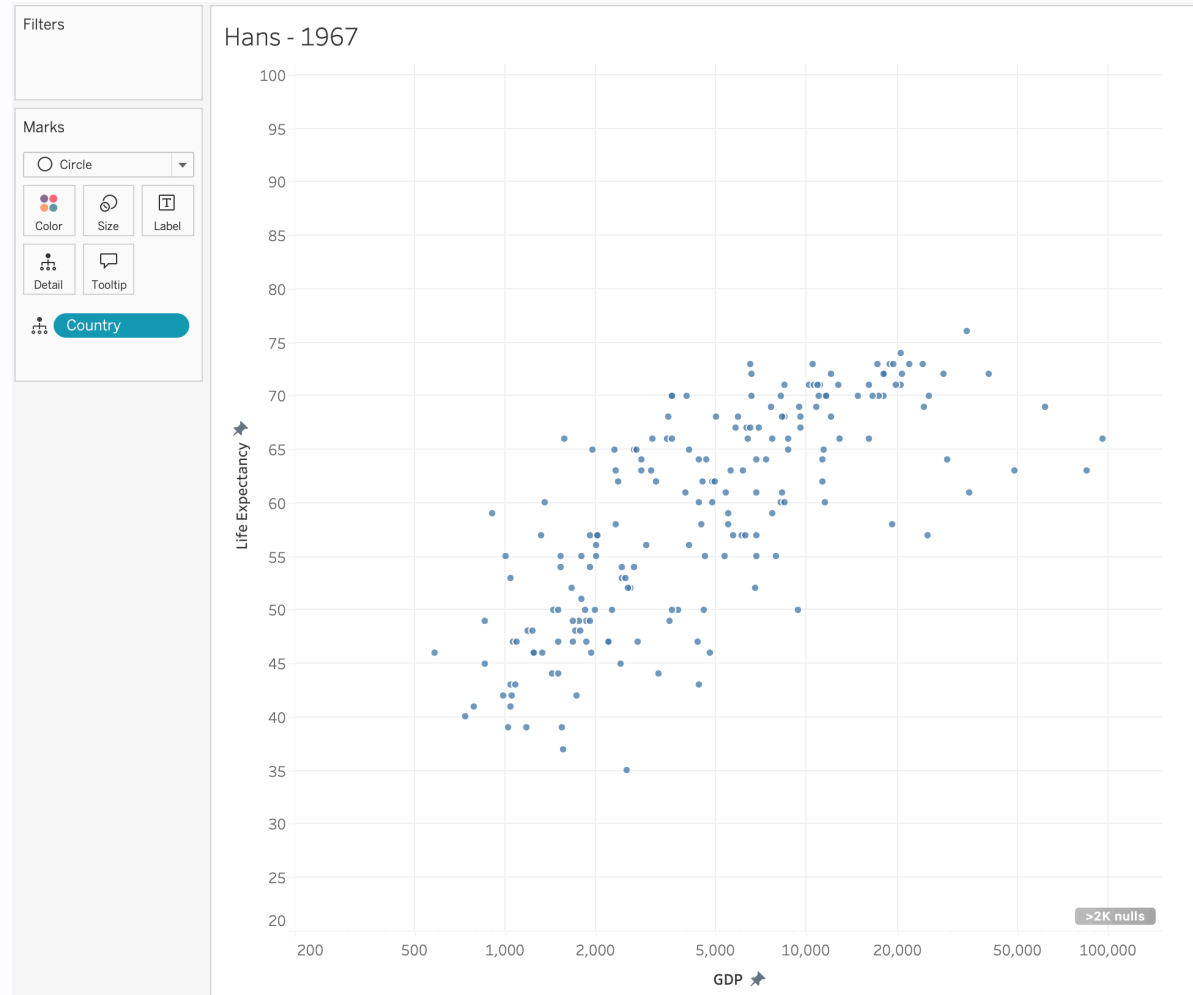
Color Size Label

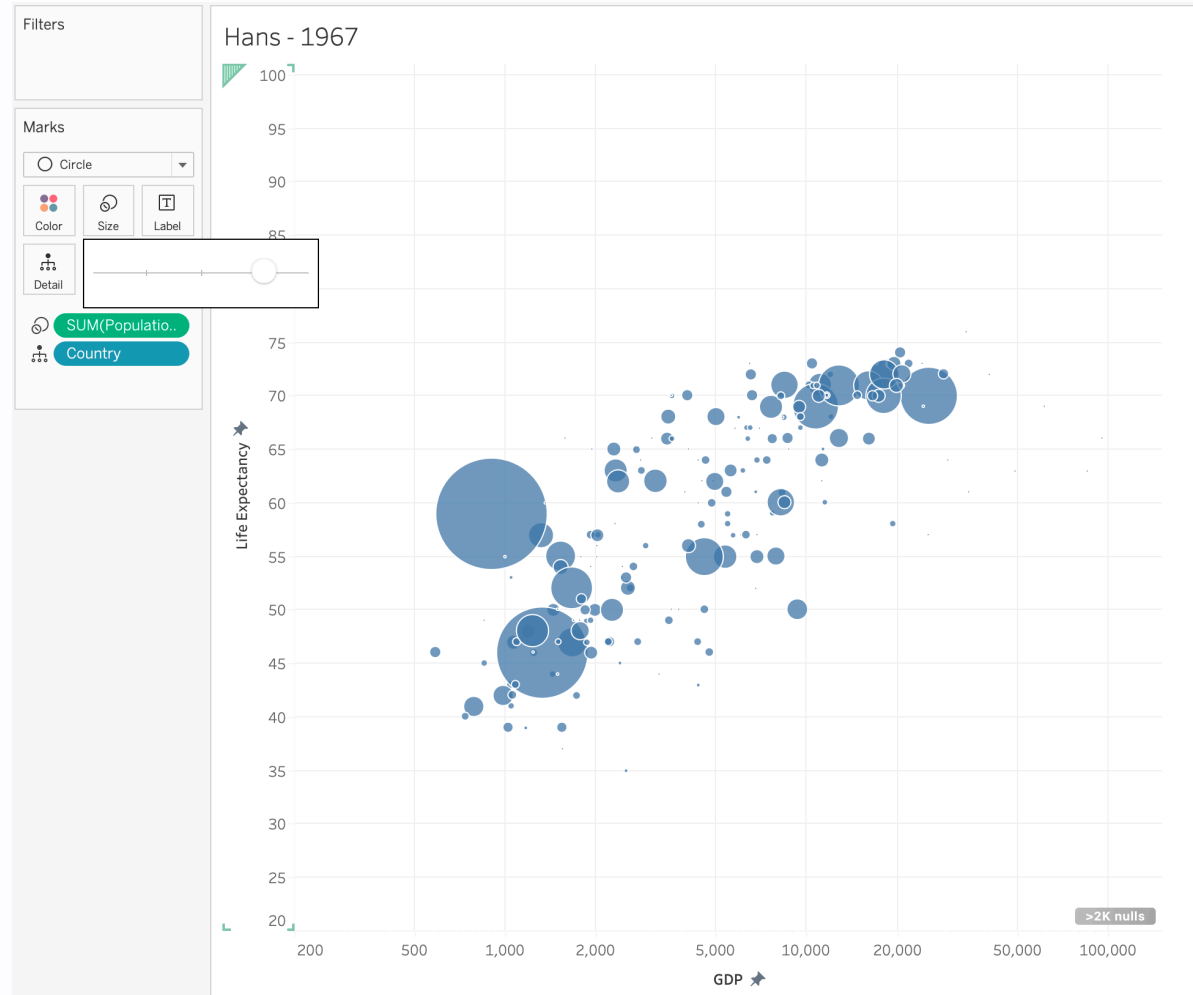
Detail Tooltip

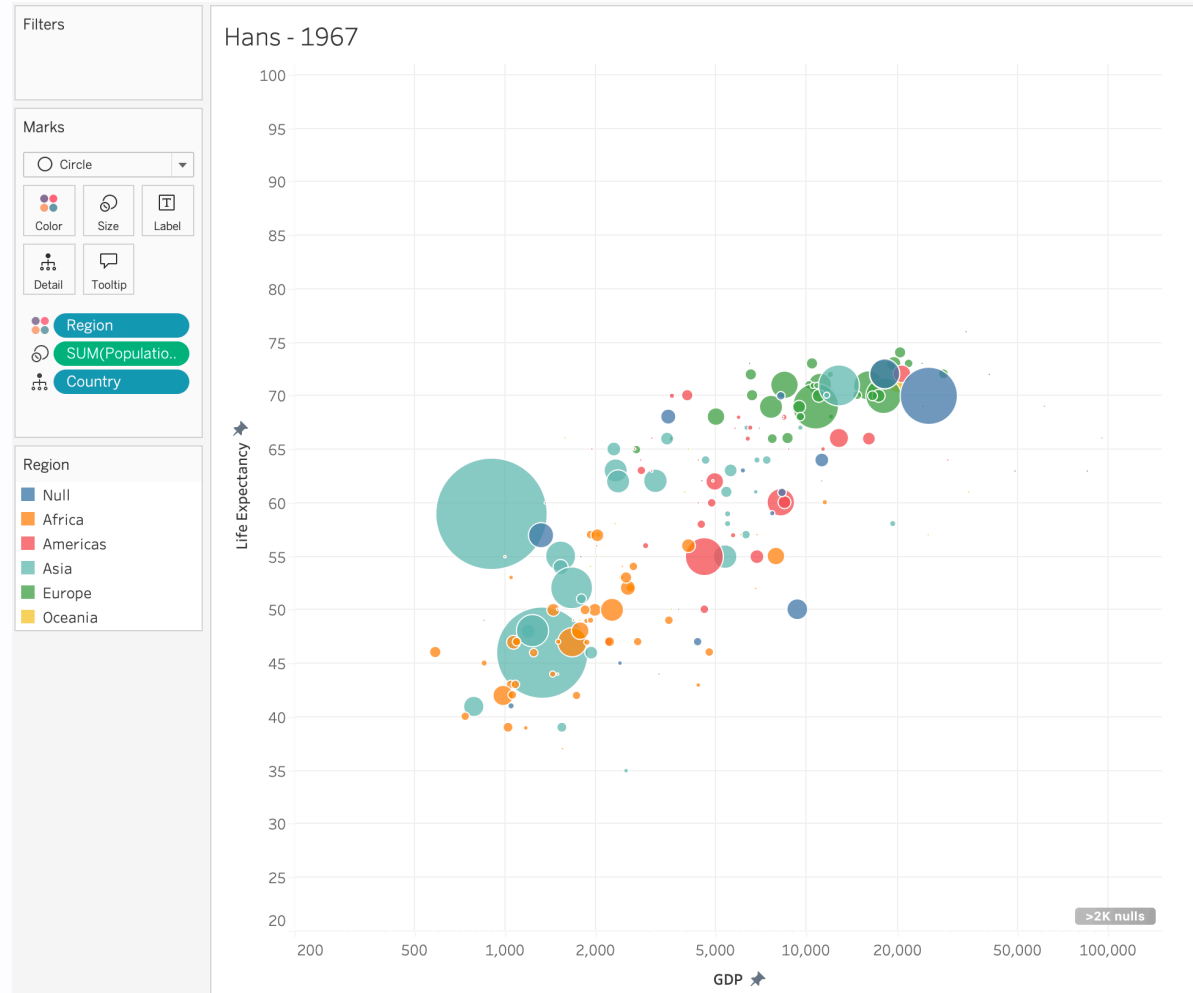
Region
SUM(Populatio..
Country
Country

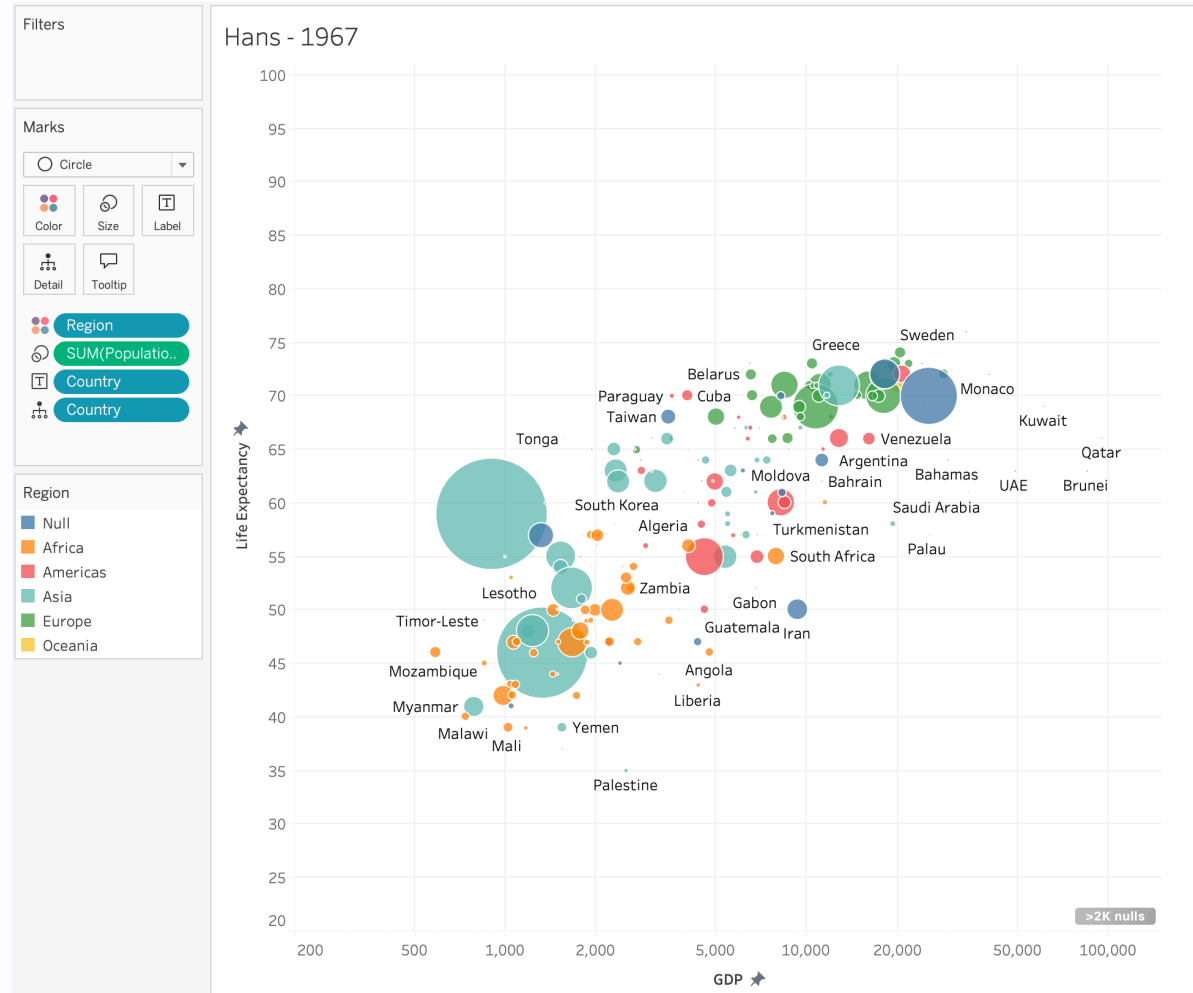
Hans - 1967

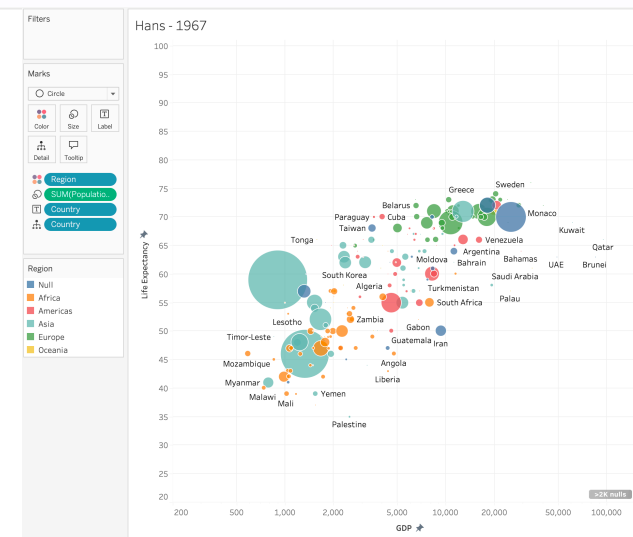
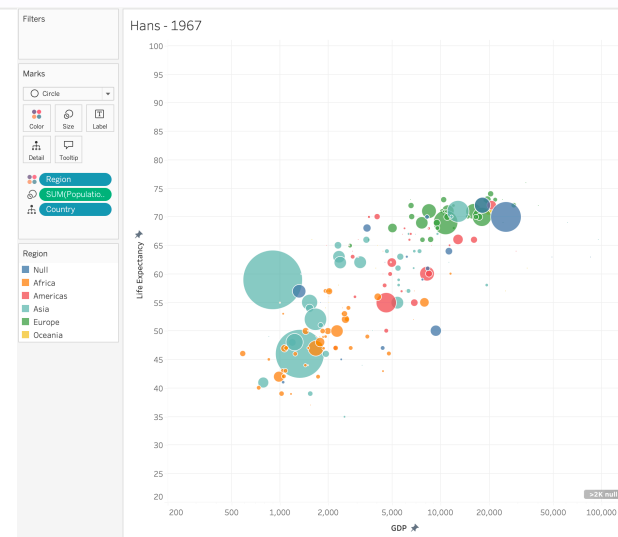
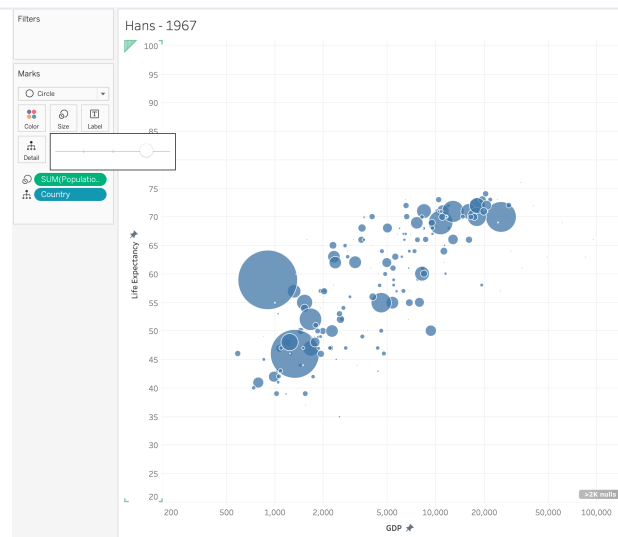
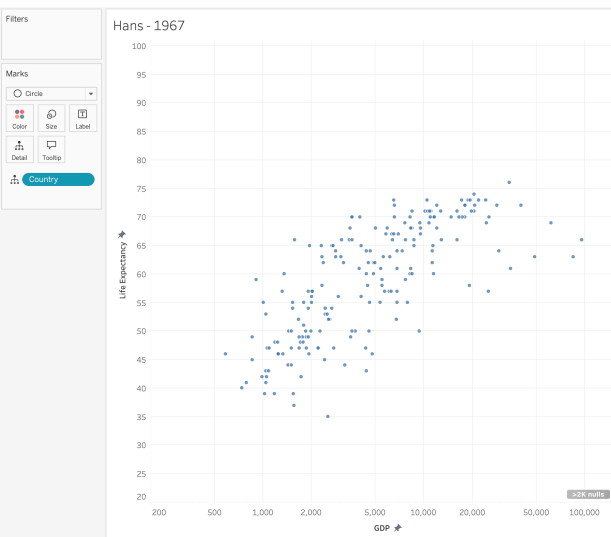












What do you see here?

<https://commons.wikimedia.org/wiki/File:Nightingale-mortality.jpg#/media/File:Nightingale-mortality.jpg>

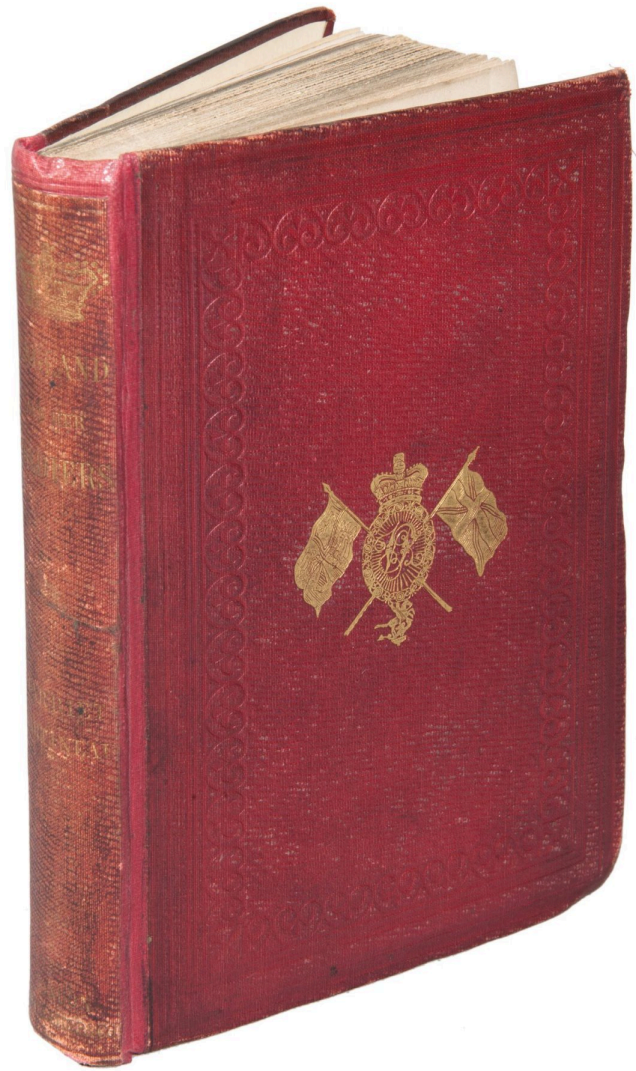


The lady with the lamp

Florence Nightingale (1820-1910)



MISS NIGHTINGALE, IN THE HOSPITAL, AT SCUTARI.—(SEE PRECEDING PAGE.)



DIAGRAM

ENGLAND AND HER SOLDIERS.

BY HARRIET MARTINEAU.

"Let me speak proudly.—Tell the Constable
We are but warriors for the working day;
Our gayness, and our gilt, are all besmirched
With rainy marching in the painful field.
There's not a piece of feather in our host
(Good argument, I hope, we shall not fly),
And time hath worn us into slovenry;
But, by the mass, our hearts are in the trim."

KING HENRY V., Act iv. Scene 3.

"I know already by your forwardness, that you have deserved rewards and
crowns; and we do assure you, on the word of a prince, they shall be duly
paid you."
QUEEN OF ENGLAND to her Soldiers, 1588.

WITH THREE ILLUSTRATIVE DIAGRAMS.

LONDON

SMITH, ELDER, & CO. 65 CORNHILL.

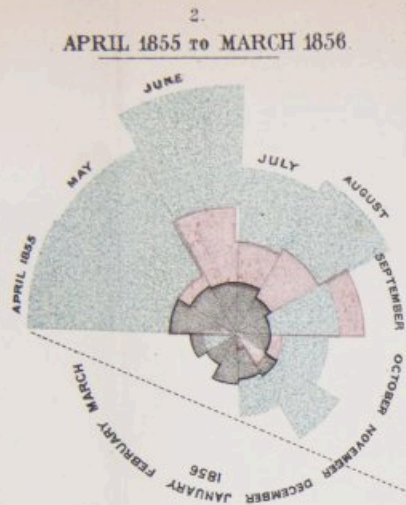
1859

The right of translation is reserved.

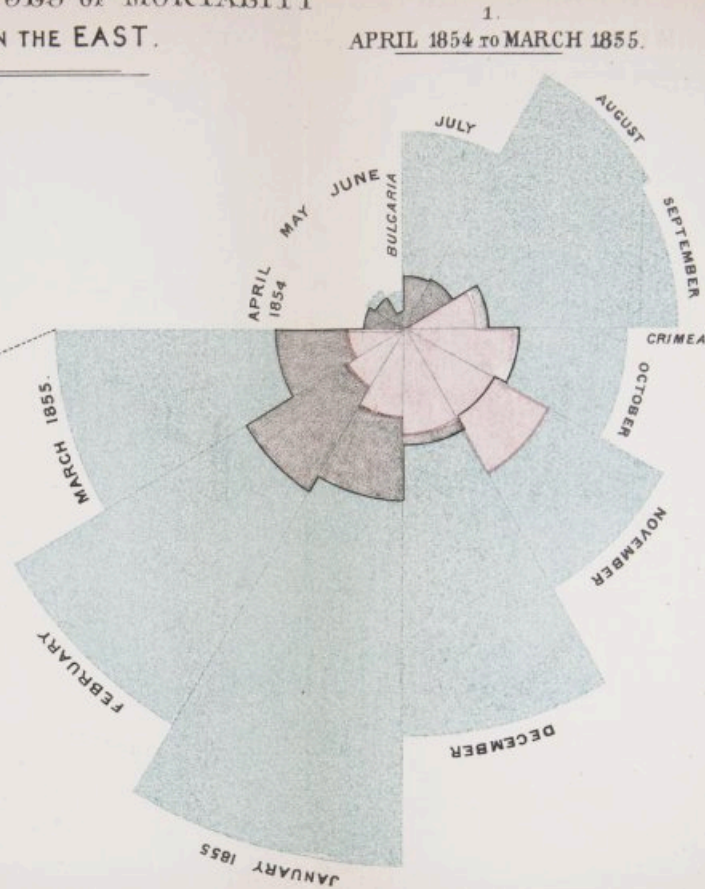
s are each me.

the circle rep
ligable Zymote
deaths from u
he deaths from
p. 1854 marks
the month.
i coincides wi
incides with t
ving the blue.

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes.
The black line across the red triangle in Nov. 1854 marks the boundary of the deaths from all other causes during the month.
In October 1854, & April 1855, the black area coincides with the red, in January & February 1856, the blue coincides with the black.
The entire areas may be compared by following the blue, the red & the black lines enclosing them.



ENGLAND AND HER SOLDIERS.

BY HARRIET MARTINEAU.

"Let me speak proudly.—Tell the Constables
We are but warriors for the working day;
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(Good argument, I hope, we shall not fly,
And time hath worn us into slowness)
But, by the mass, our hearts are in the fray."
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QUEEN OF ENGLAND to her Soldiers, 1590.

WITH THREE ILLUSTRATIVE DIAGRAMS.

LONDON
SMITH, ELDER, & CO. 65 CORNHILL.
1859

The right of translation is reserved.



Why Python?

matplotlib

 **seaborn**

 **plotly**

b**keh**

In class Python-ers

Explore python on your own (10 minutes)

Group share (10 minutes)

[Python Notebook Example](#)

Week 5 課題